

DDR5 SPD EEPROM with Hub and Integrated Temperature Sensor (SPD5)

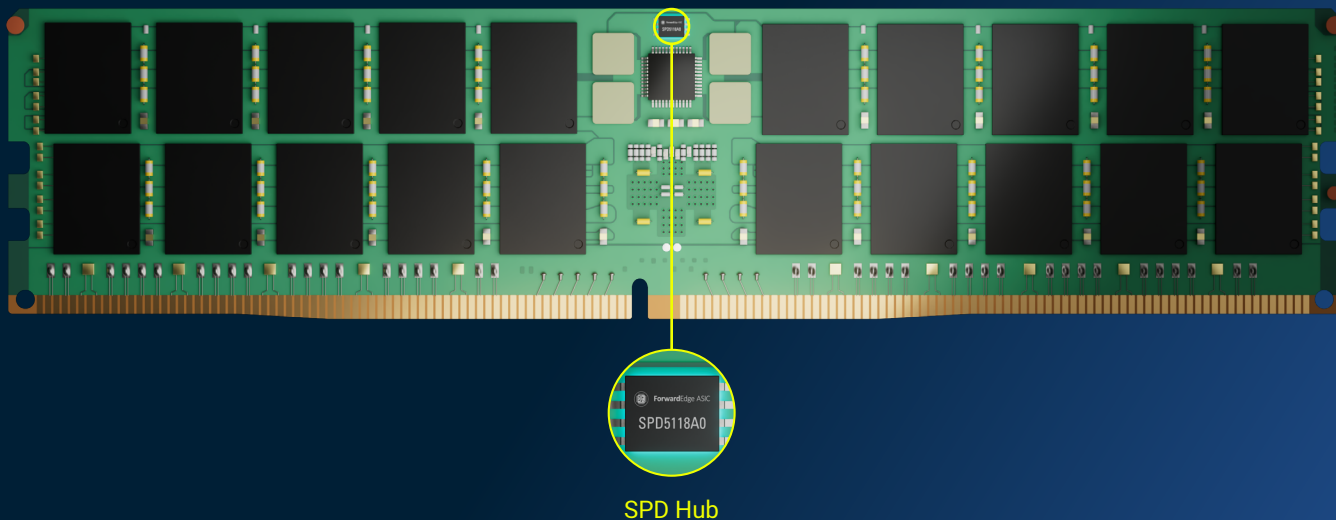
Product Description

SPD5118A0 is a DDR5 Serial Presence Detect (SPD5) Hub device which contains 1024 bytes of non-volatile memory arranged as 16 blocks of 64 bytes per block. Each block may optionally be write-protected via software command. Write protection for each block may be overridden in an offline programmer environment while overrides are prevented in normal use.

The SPD5 Hub device operates from 1.8V nominal power supply input. The SPD5 Hub device is intended to operate up to 12.5 MHz on an I3C Basic bus or up to 1 MHz on an I2C bus. The SPD5 Hub device is intended to interface to I2C /I3C Basic buses which have multiple devices on a shared bus and must be uniquely addressed with fixed addressing on the same bus. The SPD5 Hub device responds to specific pre-defined I2C / I3C Basic device select codes on a host interface bus. The SPD5 Hub device also incorporates a second local I2C /I3C Basic bus and pass-through of commands from the host bus onto the local bus for addressing of I2C /I3C Basic devices on the local bus (Hub function). SPD5118A0 device incorporates thermal sensing capability which is controlled and read over I2C /I3C Basic bus.

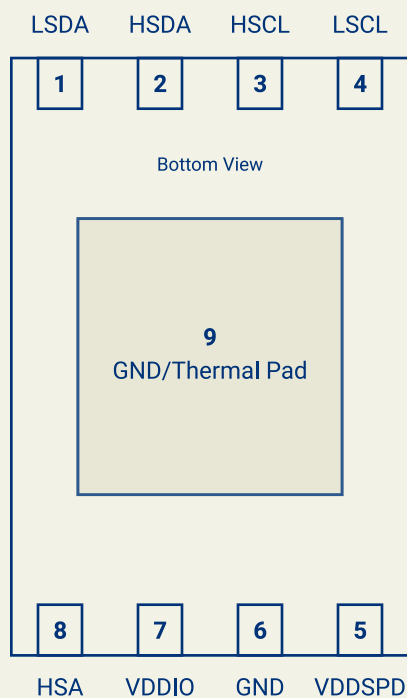
Feature List

- Two-wire I²C and I3C serial bus interface
- Single device load on the host bus
- Up to 12.5 MHz transfer rate
- 1.8 V VDDSPD and 1.0 V VDDIO power supply
- Supports 1.0V, 1.1V, 1.2V Push-Pull IO levels
- Host supports 1.0V, 1.1V, 1.2V, 1.8V, 2.5V, and 3.3V Open-drain IO levels
- 1024 bytes of non-volatile memory (16 blocks; 64 bytes per block)
- Hub function, supporting up to 8 unique addressing
- Integrated Temperature Sensor with high accuracy for JEDEC specification compliance
- Temperature Range from -40°C to +125°C
- Packet Error Check (PEC) & Parity Error Check functions
- In Band Interrupt (IBI)
- Write protection for each block of NVM
- Package: 9-contact thermally enhanced DFN
- 5-byte unique serial number to support authentication



Package Pinout

The SPD5 Hub device is packaged in a 9-contact thermally enhanced PSON-8, MO-229 Variation V/W/UCED-3. The package size is 2.0 mm x 3.0 mm.



Applications

- DDR5 DIMMs for servers (RDIMM, LRDIMM and MRDIMM)
- DDR5 DIMMs for desktops and laptops (UDIMM and SODIMM)

Contact Information

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